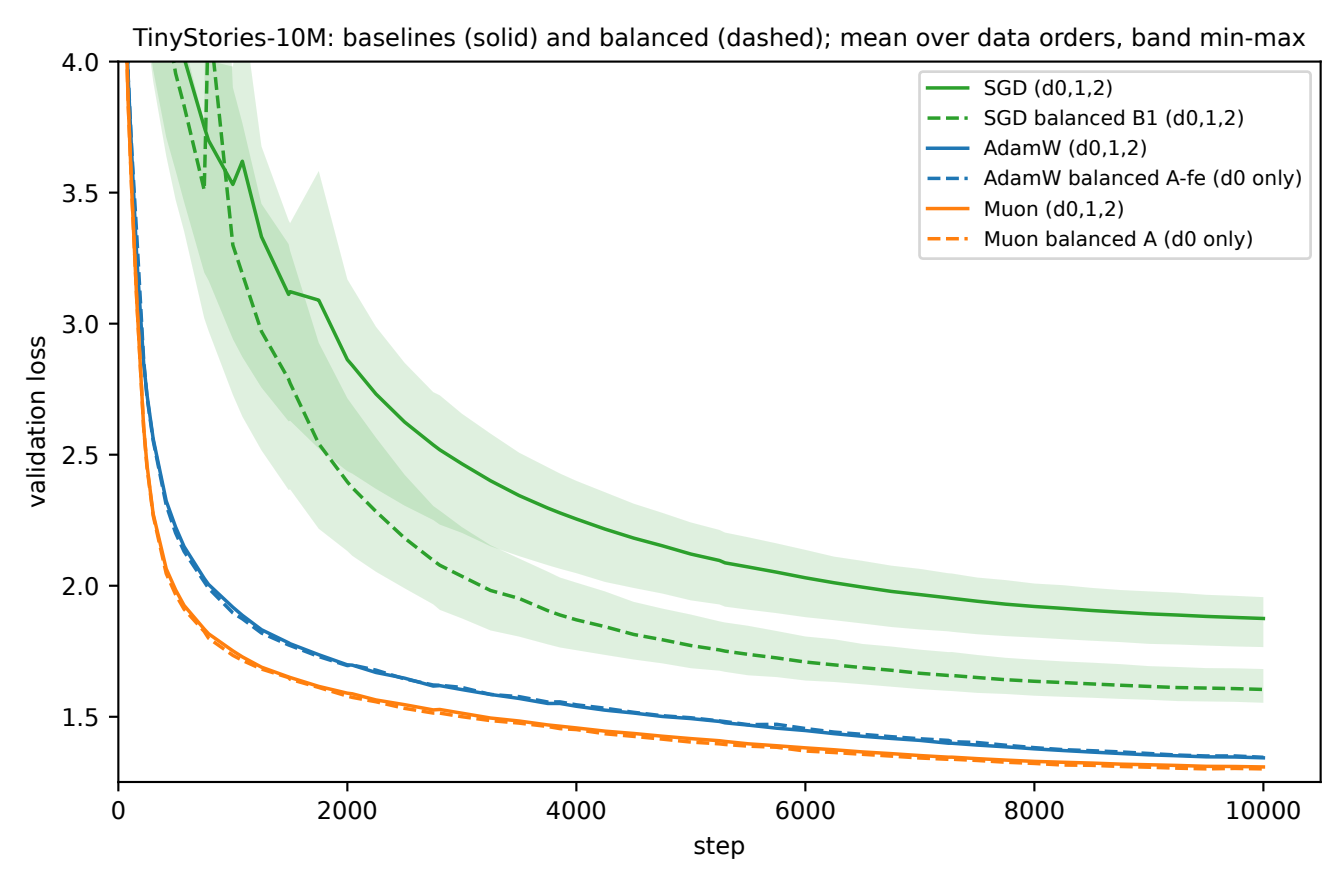
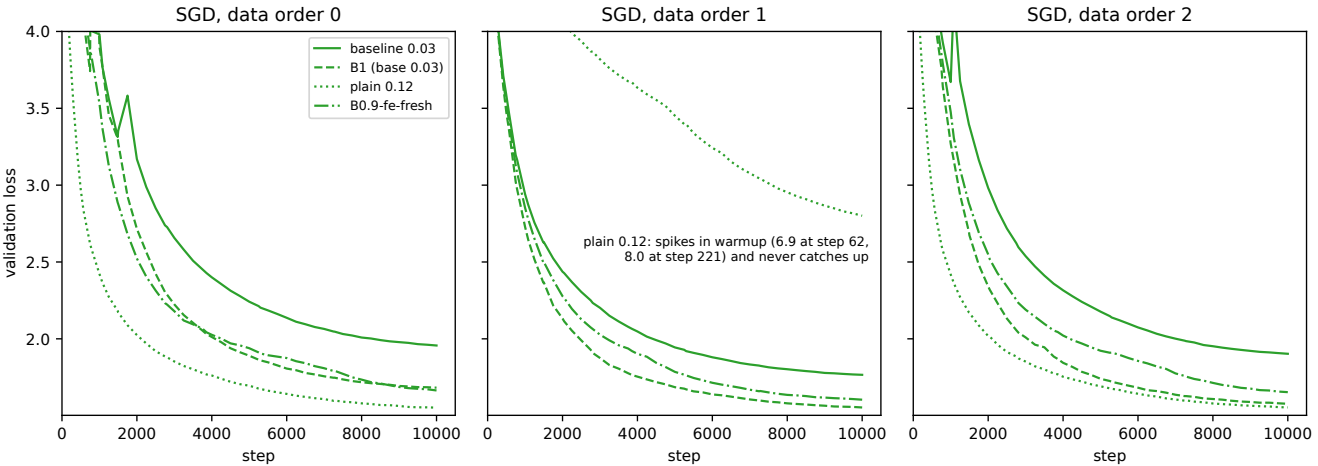
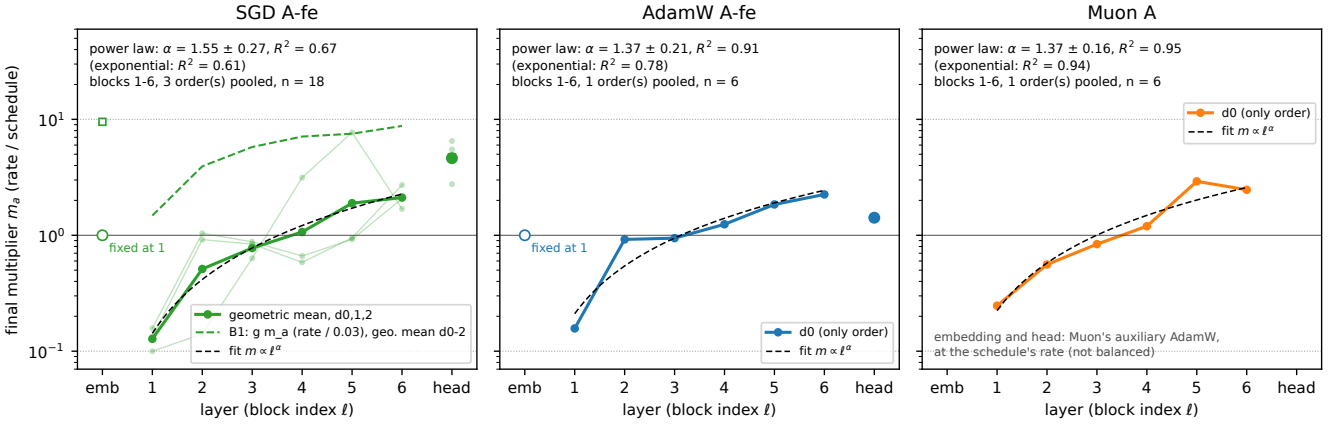
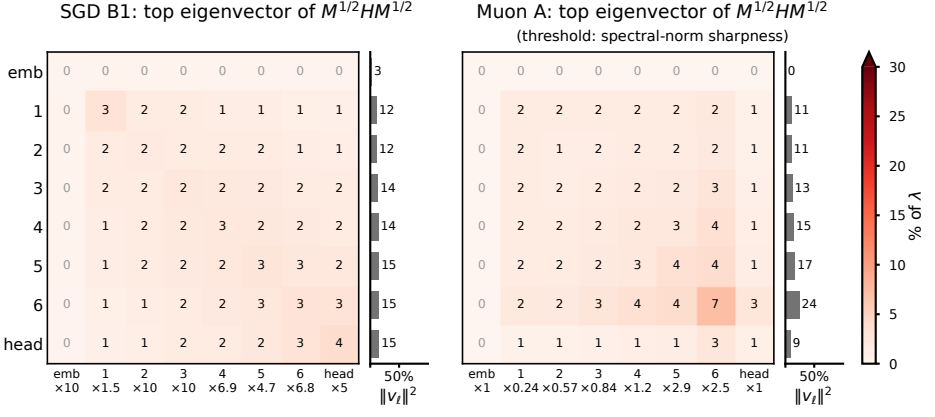

TinyStories-10M: baselines (solid) and balanced (dashed); mean over data orders, band min-max
validation loss
step
4.0
3.5
3.0
2.5
2.0
1.5
0
2000
4000
6000
8000
10000
SGD (d0,1,2)
SGD balanced B1 (d0,1,2)
AdamW (d0,1,2)
AdamW balanced A-fe (d0 only)
Muon (d0,1,2)
Muon balanced A (d0 only)


SGD, data order 0
SGD, data order 1
SGD, data order 2
validation loss
step
baseline 0.03
B1 (base 0.03)
plain 0.12
B0.9-fe-fresh
plain 0.12: spikes in warmup (6.9 at step 62,
8.0 at step 221) and never catches up
4.0
3.5
3.0
2.5
2.0
0
2000
4000
6000
8000
10000

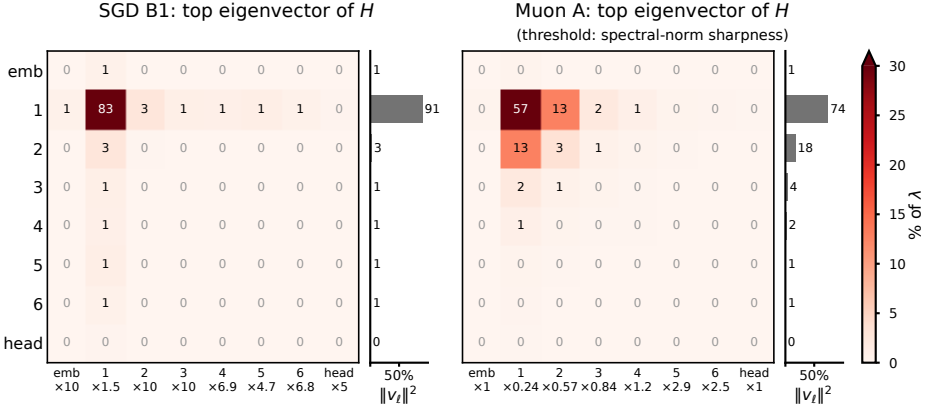


Clear trend in all three: the rate rises with depth, block 1 lowest (0.10-0.25x) and blocks 5-6 highest (about 2-3x); a power law fits better than an exponential. SGD's single orders scatter widely around it (d1: blocks 1-2 at 0.1-0.14x, block 5 at 7.7x); AdamW and Muon have one order each.

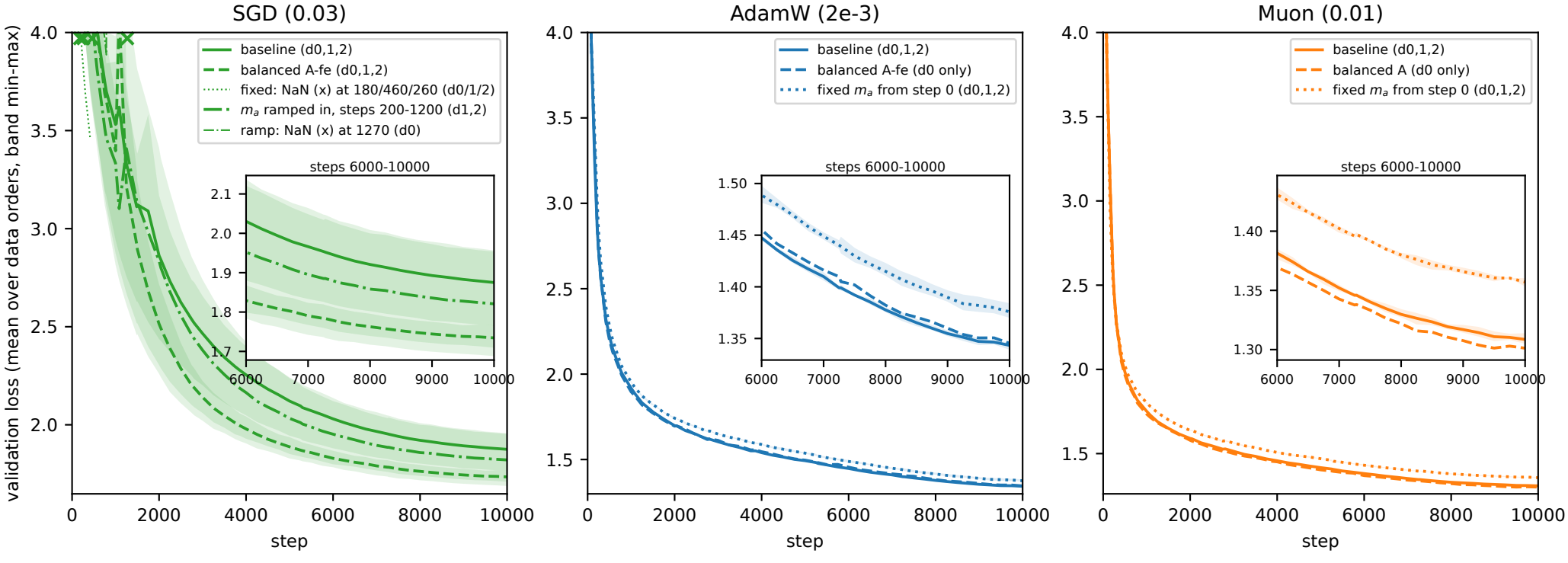
Balanced runs, mean over the checkpoints at steps 5000-10000. $M = \mathrm{diag}(m_\ell)$, with m_ℓ each layer's rate multiplier logged by the balancer (under the labels).
Row sums $\|v_\ell\|^2$ by the bars. Left out, no checkpoint with optimizer state (for P): ts10m_balanced/adamw_A-fe_d0.



Balanced runs, mean over the checkpoints at steps 5000-10000. Plain operator (without M); m_t: each layer's rate multiplier logged by the balancer (under the labels).
Row sums $\|v_t\|^2$ by the bars. Left out, no checkpoint with optimizer state (for P): ts10m_balanced/adamw_A-fe_d0.



TinyStories-10M: fixed final multipliers (dotted; no balancing, m_a of the balanced run on d0, from step 0; SGD also ramped in after warmup, dash-dot) against the balanced run (dashed) and the baseline (solid)



	emb	1	2	3	4	5	6	head
m_a	1*	0.13	0.92	0.84	0.58	0.95	2.71	6.51

	emb	1	2	3	4	5	6	head
m_a	1*	0.16	0.92	0.94	1.24	1.84	2.25	1.42

	emb	1	2	3	4	5	6	head
m_a	1*	0.25	0.56	0.84	1.20	2.92	2.47	1*

* kept at the schedule's rate: the embedding as in the -fe runs; for Muon the embedding, head and norms (its auxiliary AdamW). SGD ramp: m_a^f, f = 0 to step 200, linear to 1 at step 1200.

final validation loss per data order; speedup = 10000 / first evaluated step at which the run reaches the baseline's final loss (same order)

order	baseline	A-fe	speedup	fixed	speedup	ramp	speedup
d0	1.957	1.766	2.11x	NaN @180	-	NaN @1270	-
d1	1.766	1.709	1.38x	NaN @460	-	1.688	1.54x
d2	1.903	1.727	2.11x	NaN @260	-	1.954	never

order	baseline	A-fe	speedup	fixed	speedup
d0	1.339	1.345	never	1.370	never
d1	1.346	-	-	1.373	never
d2	1.345	-	-	1.384	never

order	baseline	A	speedup	fixed	speedup
d0	1.305	1.301	1.08x	1.353	never
d1	1.307	-	-	1.358	never
d2	1.314	-	-	1.360	never

AdamW with fixed multipliers (no balancing; adamw_fixed_d2), mean over the checkpoints at steps 5000, 7500 and 10000.
$M = \mathrm{diag}(m_\ell)$, with m_ℓ each layer's fixed rate multiplier (under the labels): the final ones of adamw_A-fe_d0. Row sums $\|v_\ell\|^2$ by the bars.

